Boolean conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected Boolean-condition record.

Explain comparisons, trace sub-results, find an incorrect connector and state a validation limit.

Keep one habit: Keep written rule, supplied input types, individual booleans, final condition and observed evidence together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Expressions and operator order

> **My remembered idea, example and correction**

From a paper algorithm to Python

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 5

W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Activity contract: nonnegative integer minutes and Boolean quiet; suitable=(minutes<=12) and quiet. W2 (11,True)→time True/suitable True; (12,False)→True/False; (13,True)→False/False. Exact 12 is included. Separate age interval 12..15 uses both inclusive checks; age 11 fails lower, 16 fails upper. OR is inclusive for Boolean inputs, and every equality alternative must be written.

  None of these expressions validates arbitrary input or real-world permission.

- Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Activity contract: nonnegative integer minutes and Boolean quiet; suitable=(minutes<=12) and quiet. W2 (11,True)→time True/suitable True; (12,False)→True/False; (13,True)→False/False. Exact 12 is included. Separate age interval 12..15 uses both inclusive checks; age 11 fails lower, 16 fails upper. OR is inclusive for Boolean inputs, and every equality alternative must be written.

  None of these expressions validates arbitrary input or real-world permission.

- For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Activity contract: nonnegative integer minutes and Boolean quiet; suitable=(minutes<=12) and quiet. W2 (11,True)→time True/suitable True; (12,False)→True/False; (13,True)→False/False. Exact 12 is included. Separate age interval 12..15 uses both inclusive checks; age 11 fails lower, 16 fails upper. OR is inclusive for Boolean inputs, and every equality alternative must be written.

  None of these expressions validates arbitrary input or real-world permission.

- Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Activity contract: nonnegative integer minutes and Boolean quiet; suitable=(minutes<=12) and quiet. W2 (11,True)→time True/suitable True; (12,False)→True/False; (13,True)→False/False. Exact 12 is included. Separate age interval 12..15 uses both inclusive checks; age 11 fails lower, 16 fails upper. OR is inclusive for Boolean inputs, and every equality alternative must be written.

  None of these expressions validates arbitrary input or real-world permission.

- Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Predict all four **boolean**s. Eligibility requires science AND at most100 pages AND availability. Repeat with genre="history" and all other values unchanged. Which OR result could wrongly suggest the genre requirement passed?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Combine requirements without replacing AND

```
pages = 90
genre = "science"
available = False
print(genre == "science" and pages <= 100)
print((genre == "science" and pages <= 100) and available)
print(not available)
print(genre == "science" or pages <= 100)
```

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled **boolean** conditions **evidence** sheet to Python readiness portfolio. Show how you can translate stated requirements into **comparison**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Trace separate checks for **Boolean** conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False).

  Choose a case that detects replacing and with or between the range check and available.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

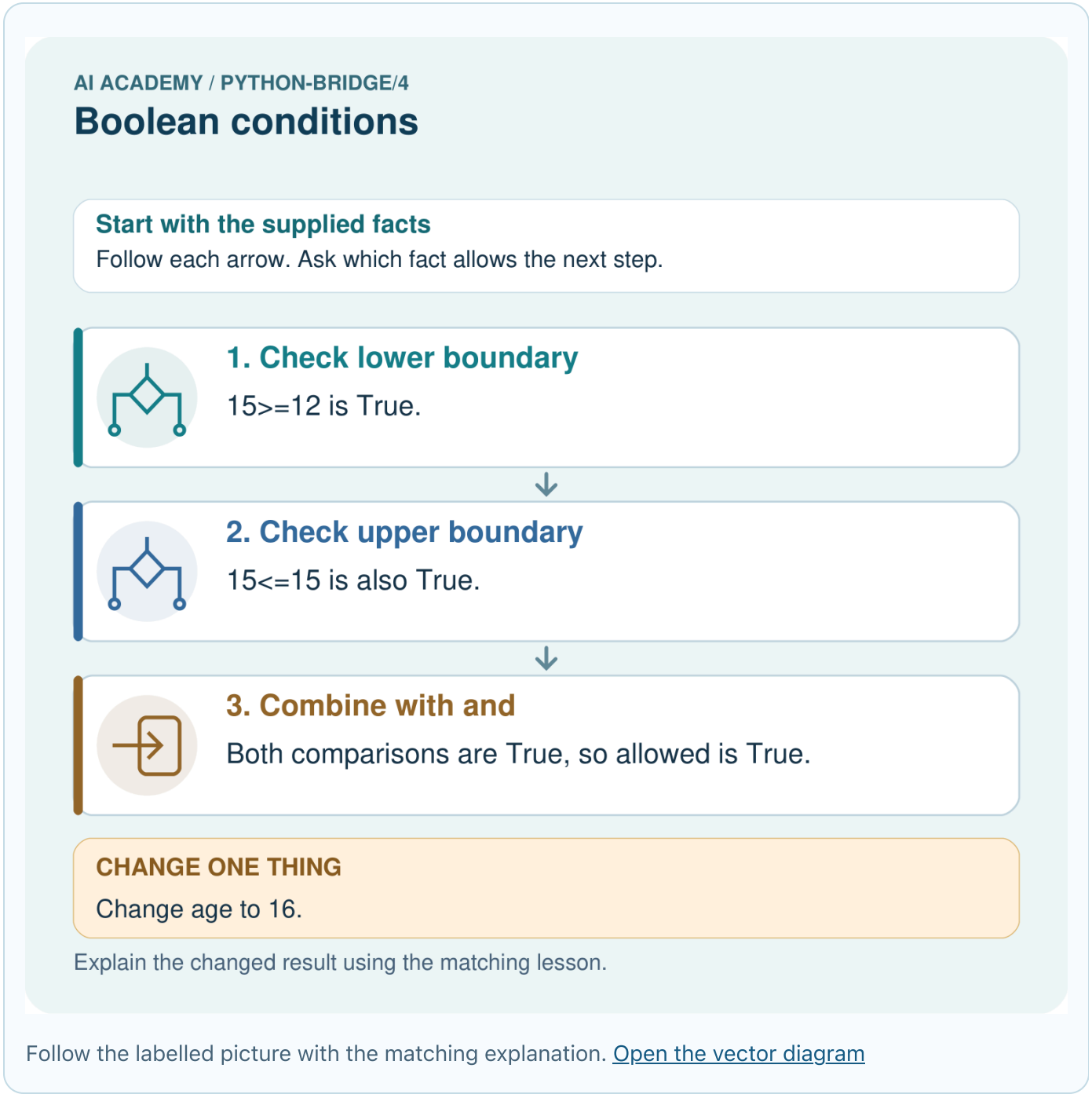
- Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

- Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

- Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.
- Which labelled step matches this detail? Both **comparison**s are True, so allowed is True.
- 1. Check lower boundary
- 2. Check upper boundary
- 3. Combine with and

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Combine with and Both **comparison**s are True, so allowed is True. A **Boolean** condition represents a yes-or-no result.

And requires both conditions; checking only one boundary would accept values outside the intended interval.

Changed-case question and hint

- Changed-case practice: Change age to 16. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The result becomes False.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The result becomes False. The lower condition remains True, but the upper condition fails.

Repair and keep the record

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Boolean conditions: the case

AI ACADEMY / PYTHON-BRIDGE/4 · 01 · LOOK AT THE FACTS

Boolean conditions: the case

The situation

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Supplied evidence

Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

Trace boolean conditions and explain the changed program before running it.

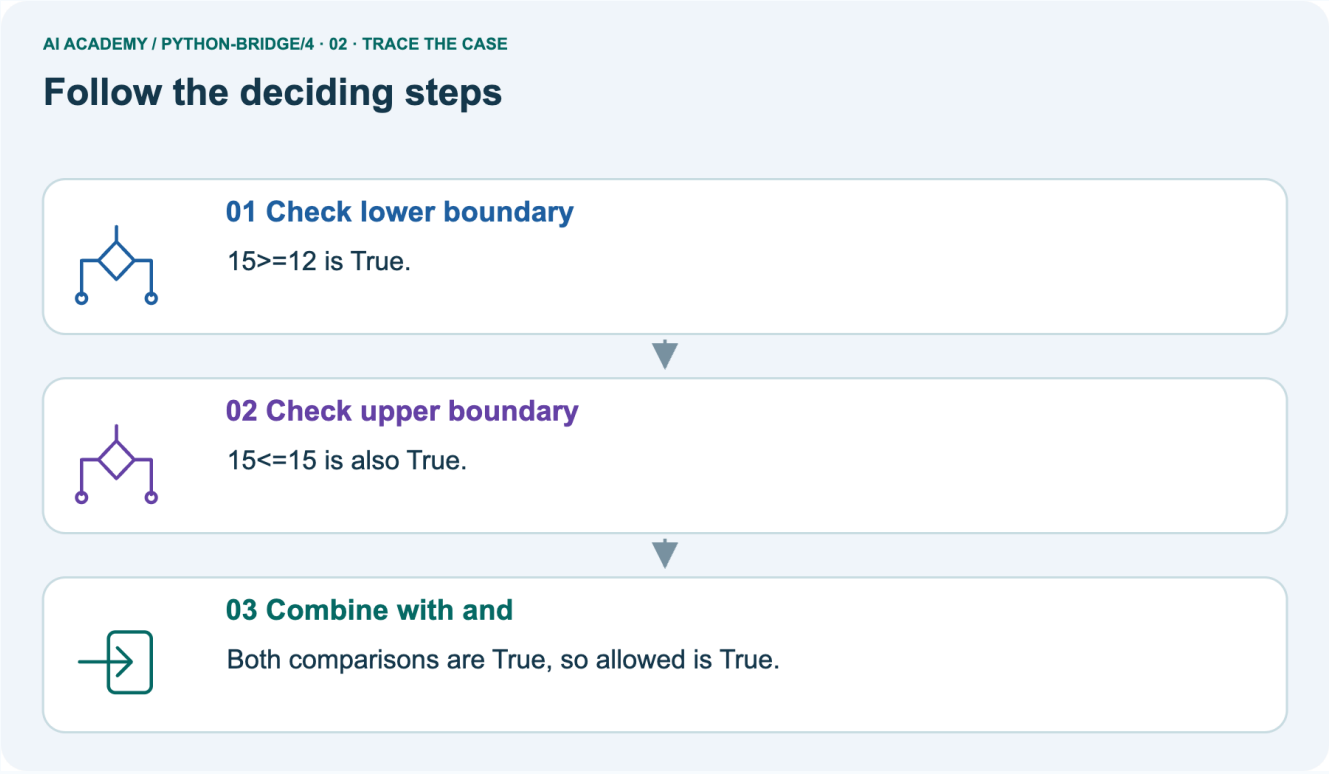
Follow the labelled picture with the matching explanation. Open the vector diagram

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.
Supplied facts: Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check lower boundary: 15>=12 is True. → Check upper boundary: 15<=15 is also True. → Combine with and: Both **comparison**s are True, so allowed is True.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two requirements must both hold

AI ACADEMY / PYTHON-BRIDGE/4 · 03 · INSPECT THE LESSON MODEL

Two requirements must both hold

Input case	Time check	Combined result
12 min, quiet False	12 <= 12 is True	True and False is False
12 min, quiet True	12 <= 12 is True	True and True is True
15 min, quiet True	15 <= 12 is False	False and True is False

Passing only one requirement does not make the activity suitable.

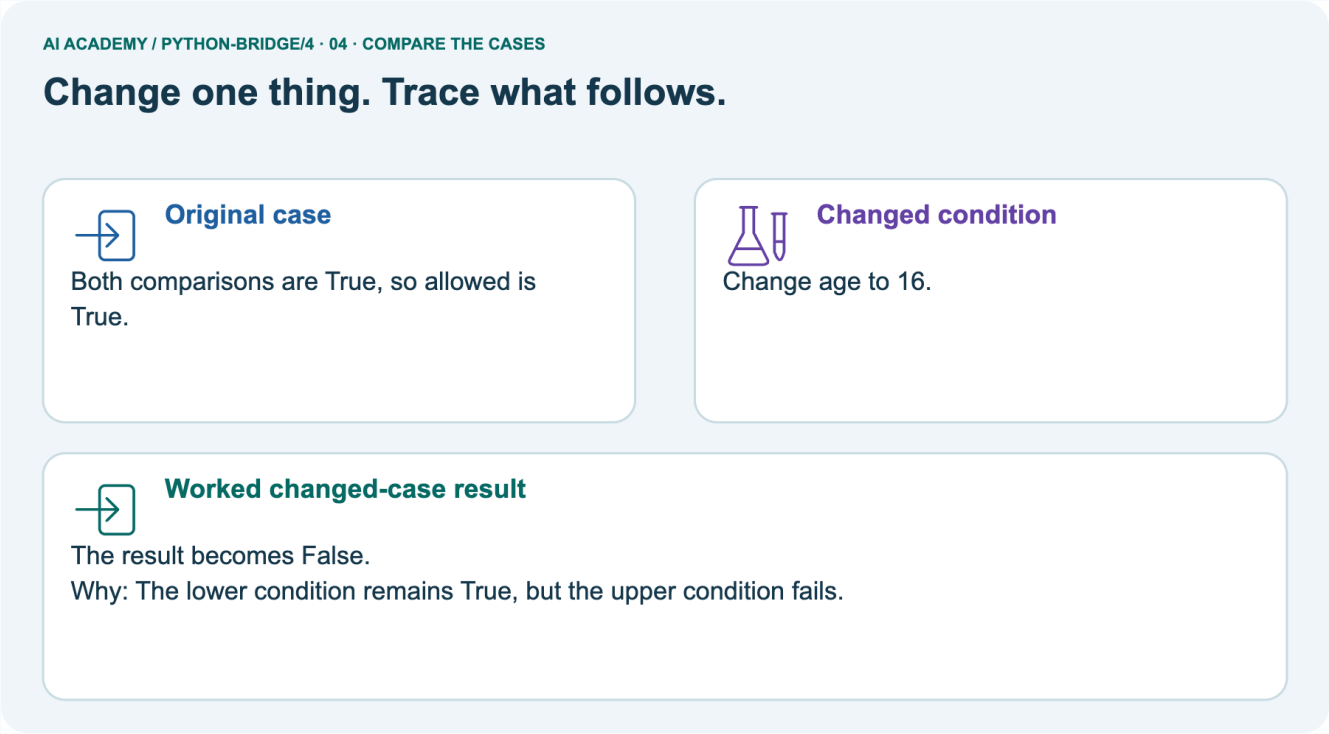
Follow the labelled picture with the matching explanation. Open the vector diagram

Two requirements must both hold. 12 min, quiet False | 12 <= 12 is True | True and False is False. 12 min, quiet True | 12 <= 12 is True | True and True is True. 15 min, quiet True | 15 <= 12 is False | False and True is False

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



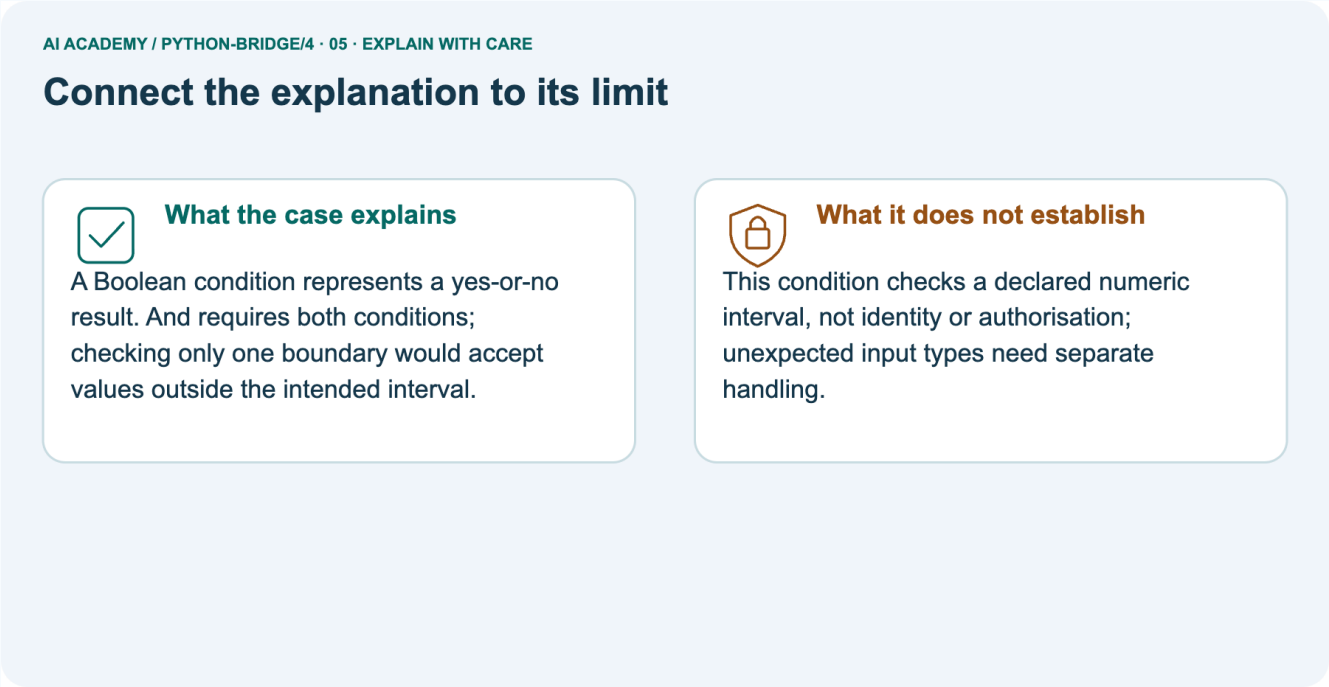
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both **comparison**s are True, so allowed is True. Changed condition: Change age to 16. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

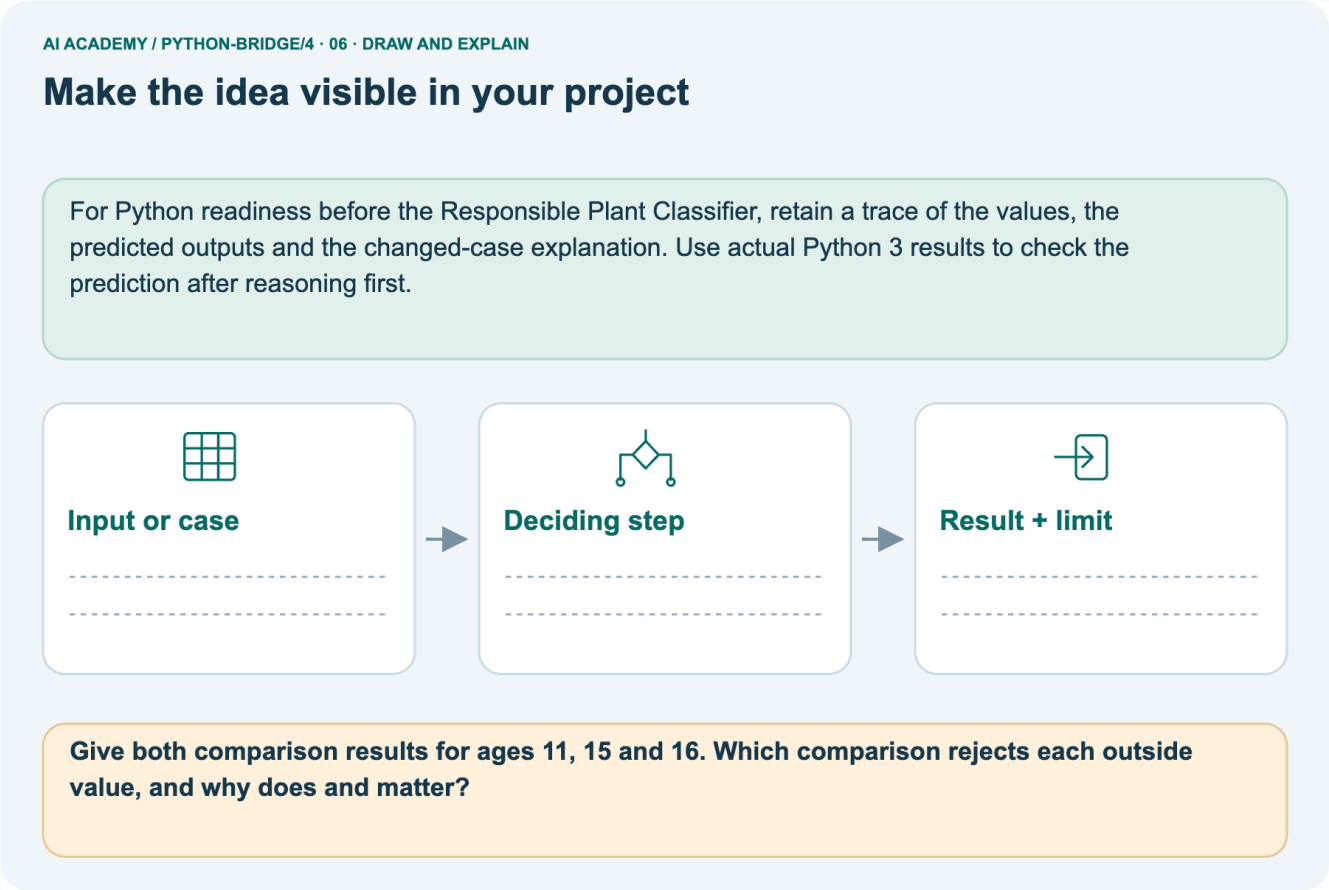
A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Limit: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Boolean conditions: the case

02. Follow the deciding steps

03. Two requirements must both hold

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.